Results pack Q3: Cumulative fluid balance and ventilator-free days

FLUID-ICU is a SIMULATED dataset made for teaching. No real patients. Neudata | Foundations of Clinical Research, Level 1

SIMULATED DATA: NOT REAL PATIENTS. FLUID-ICU was generated by computer for this course. Write your manuscript as a training exercise and state clearly that the data are simulated.

1. The question

In adults with sepsis in the ICU, is a higher cumulative fluid balance by day 3 associated with fewer ventilator-free days to day 28?

2. Design and data

- Design: multicentre prospective cohort study (observational).
- Setting: 4 ICUs (ICU A-D), consecutive adults with sepsis, enrolment January 2023 to December 2024.
- Exposure: cumulative fluid balance at day 3 per 10 ml/kg (continuous); secondary: positive vs negative.
- Outcome: ventilator-free days (VFD) to day 28: days alive and free of mechanical ventilation; patients who died = 0; never ventilated and alive = 28 (vent_free_days_28).
- Analysis sample: 618 patients; adjusted models 527 complete cases; 313 were ventilated on admission.
- Pre-specified confounders (SAP): age, SOFA score, lactate, vasopressor, mechanical ventilation, chronic kidney disease.
- Software: R 4.6 (packages gtsummary, survival, mice). Scripts: Course/Data/R/.

Participant flow: 620 patients enrolled -> 2 without a day-3 fluid balance (weight or output missing after cleaning) -> 618 in the exposure analysis -> 91 without lactate (or age) -> 527 in the complete-case adjusted model.

3. Table 1: who was studied

Table 1. Baseline characteristics by fluid-balance group

Characteristic	Overall N = 618[1]	Negative balance N = 245[1]	Positive balance N = 373[1]
Age, years	62.6 (15.2)	64.6 (15.0)	61.4 (15.2)
Missing	1	1	0
Sex			
Female	243 (39%)	102 (42%)	141 (38%)
Male	375 (61%)	143 (58%)	232 (62%)
Weight, kg	64.7 (11.9)	65.1 (12.4)	64.4 (11.6)
Diabetes	158 (26%)	70 (29%)	88 (24%)
Chronic kidney disease	64 (10%)	28 (11%)	36 (9.7%)
Heart failure	97 (16%)	54 (22%)	43 (12%)
Hypertension	282 (46%)	124 (51%)	158 (42%)
Source of infection			

Characteristic	Overall N = 618[1]	Negative balance N = 245[1]	Positive balance N = 373[1]
Respiratory	267 (43%)	101 (41%)	166 (45%)
Urinary	126 (20%)	54 (22%)	72 (19%)
Abdominal	106 (17%)	46 (19%)	60 (16%)
Skin/soft tissue	57 (9.2%)	18 (7.3%)	39 (10%)
Other	62 (10%)	26 (11%)	36 (9.7%)
SOFA score	7 (5, 9)	6 (4, 8)	8 (6, 10)
APACHE II score	19 (15, 24)	17 (12, 21)	20 (17, 25)
Missing	6	3	3
Lactate, mmol/L	2.6 (1.7, 3.7)	2.2 (1.4, 3.0)	2.9 (1.9, 4.0)
Missing	91	50	41
Mean arterial pressure, mmHg	73 (66, 80)	76 (70, 84)	71 (65, 78)
Missing	4	1	3
Vasopressor on admission	266 (43%)	61 (25%)	205 (55%)
Mechanical ventilation	313 (51%)	97 (40%)	216 (58%)

[1]Mean (SD); n (%); Median (Q1, Q3)

Mean (SD); n (%); median (Q1, Q3). Missing = number of patients without that value. Table 1 describes the patients at ICU admission only. The outcomes, such as 28-day death, are in the main results.

4. Main results

- VFD median (IQR): positive balance 20 (0 to 28) vs negative 28 (22 to 28) days; Mann-Whitney p <0.001.
- Linear regression, crude: -1.48 (95% CI -1.84 to -1.13) days per 10 ml/kg.
- Linear regression, adjusted: **-0.61 (95% CI -1.00 to -0.21) days per 10 ml/kg**, p 0.003.
- Positive vs negative balance, adjusted difference: -2.78 (95% CI -4.76 to -0.80) days.

Table 2. Ventilator-free days by fluid-balance group

Summary	Negative balance	Positive balance
n	245	373
VFD, median (IQR)	28 (22 to 28)	20 (0 to 28)
VFD, mean (SD)	21.6 (10.5)	15.2 (12.4)
VFD = 0 (died or ventilated 28 d), n (%)	45 (18%)	137 (37%)
VFD = 28 (never ventilated, alive), n (%)	133 (54%)	117 (31%)
Ventilated patients: VFD median (IQR)	23 (0 to 25)	12 (0 to 22)

Table 3. Association between fluid balance and ventilator-free days

Model	n	Difference in VFD, days (95% CI)	p
Balance per 10 ml/kg, crude (linear)	618	-1.48 (-1.84 to -1.13)	<0.001
Balance per 10 ml/kg, adjusted (linear)	527	-0.61 (-1.00 to -0.21)	0.003
Positive vs negative balance, adjusted (linear)	527	-2.78 (-4.76 to -0.80)	0.006
Positive vs negative balance, Mann-Whitney (Hodges-Lehmann)	618	-3.0 (-5.0 to 0.0)	<0.001

Negative = fewer ventilator-free days. Adjusted for age, SOFA, lactate, vasopressor, ventilation, CKD.

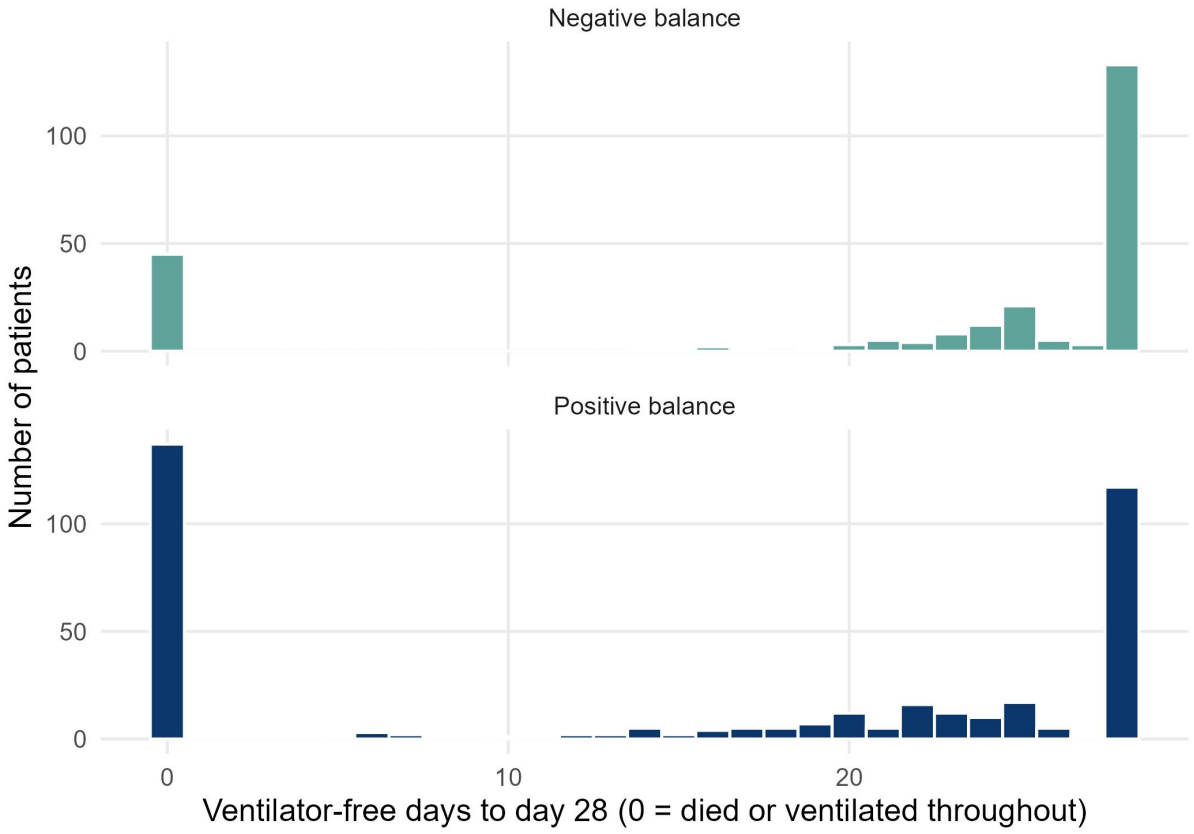


Figure 1. Distribution of ventilator-free days by group (note the two spikes at 0 and 28)

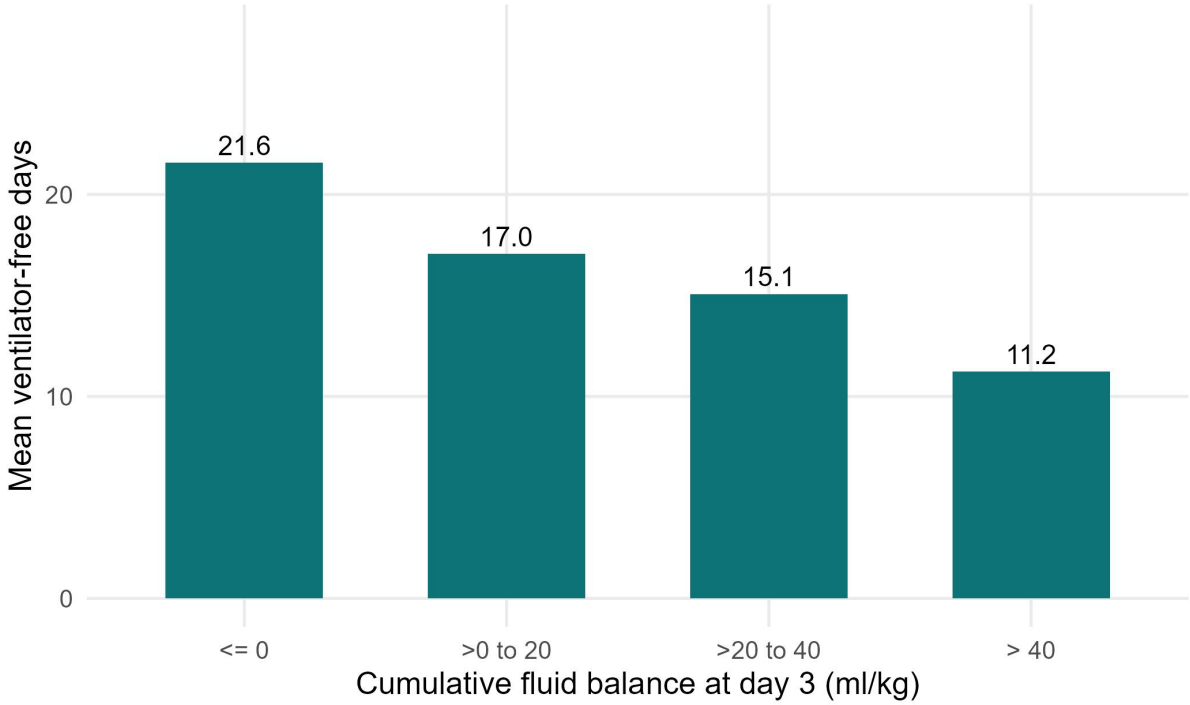


Figure 2. Mean ventilator-free days by category of cumulative fluid balance

5. What this means (plain language)

- Patients with more fluid on board had fewer days alive and off the ventilator.
- After adjustment, each extra 10 ml/kg was associated with about 0.6 fewer ventilator-free days (95% CI 0.2 to 1.0).
- VFD is not a normal-shaped number: many patients are at 0 (died) or 28 (never ventilated). The median and Mann-Whitney test describe it honestly; the linear model gives an easy-to-read average difference but its assumptions are strained.

6. What this does NOT mean

- It does NOT mean fluid keeps patients on the ventilator: VFD mixes death and ventilation time, and sicker patients get more fluid.
- A difference in mean VFD is not the same as 'each patient was ventilated one day longer'.

7. Limitations to discuss (hints)

- Composite outcome: a VFD of 0 can mean early death or 28 days on the ventilator. Say which drives the result.
- Skewed, bounded outcome: linear regression is approximate; consider the Mann-Whitney result or a proportional-odds model as sensitivity.
- Residual confounding by severity; respiratory infections both raise ventilation and may change fluid practice.
- Complete-case analysis (lactate missing).

Which STROBE items this pack helps you write

- Item 7: define the composite outcome VFD precisely
- Item 12: methods (why non-parametric and linear models)
- Item 14: descriptive data (Table 1)
- Item 15-16: outcome summaries and adjusted differences with 95% CI
- Item 19: limitations of a composite, skewed outcome